Google Developer Group
Universitas Sriwijaya

Learning 6

Advanced State Patterns & Controlled Forms

Synchronizing Components and Controlled Inputs

What We Learned Previously

- *useState*
- Updating state
- Event handling
- Conditional rendering with state
- Reactive UI

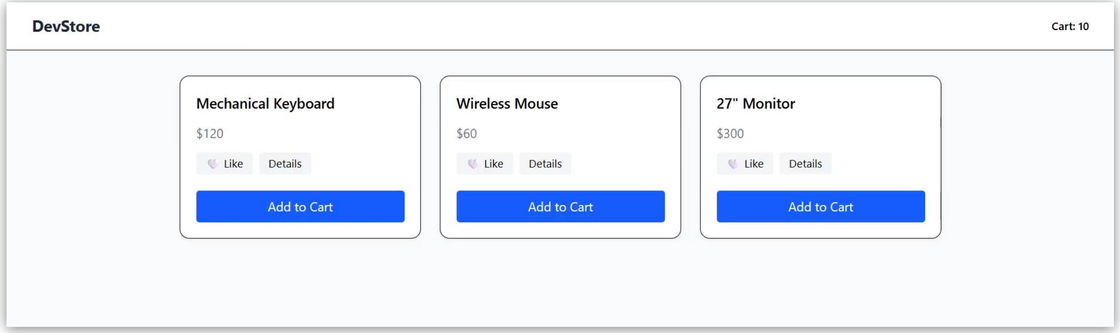
But what if multiple components need the same state?

The Real Problem

When Components Need the Same State

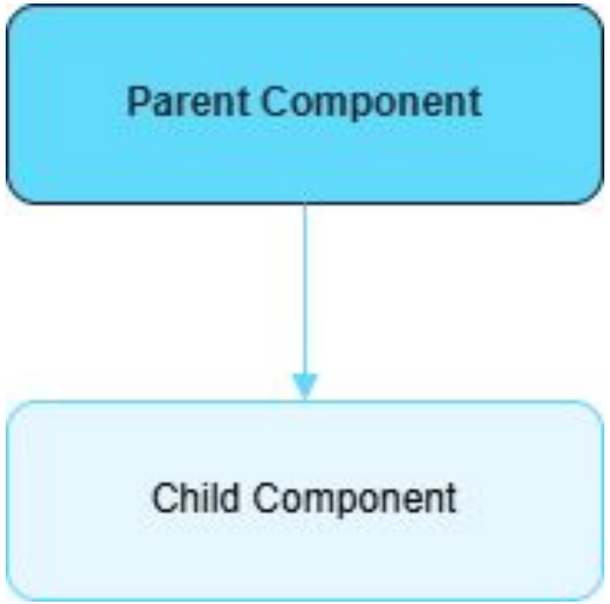
- Components: Navbar, ProductCard
- Navbar needs cart count
- ProductCard needs to update cart

Where should the state live?



React Data Flow

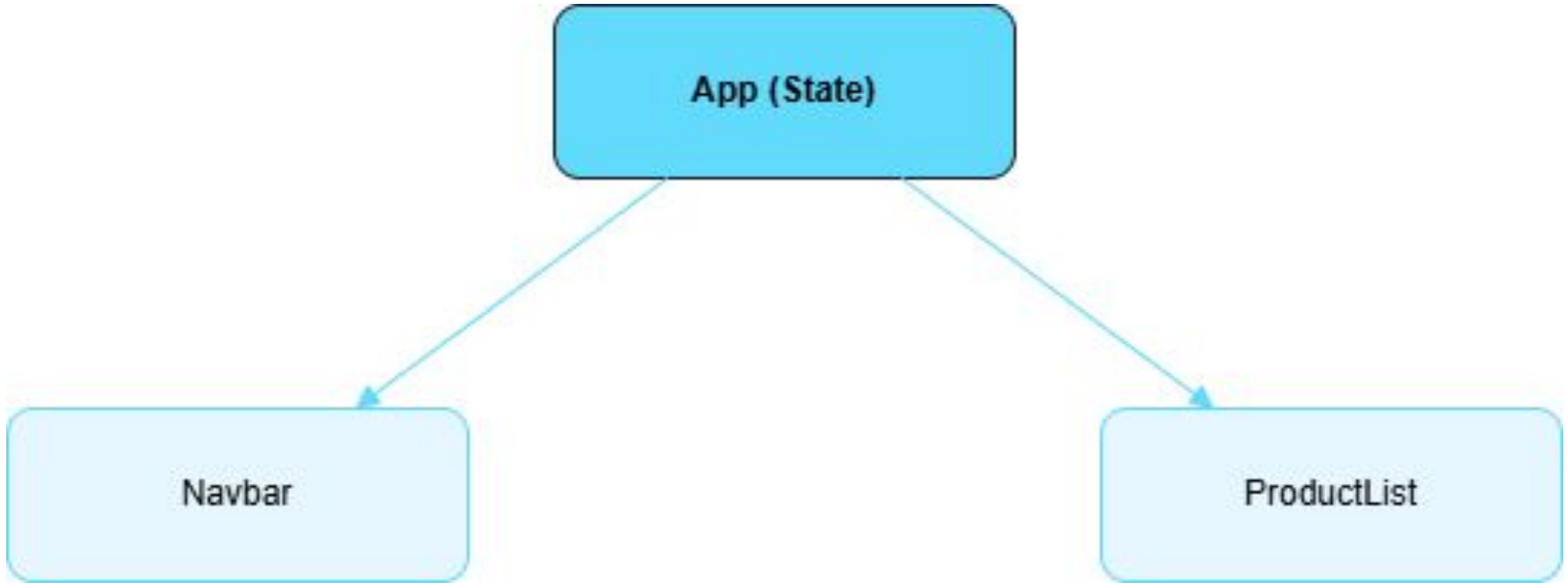
React Uses One-Way Data Flow



Data flows from parent to child via props.

The Solution - Lifting State Up

Move state to the closest common parent so it can be shared.

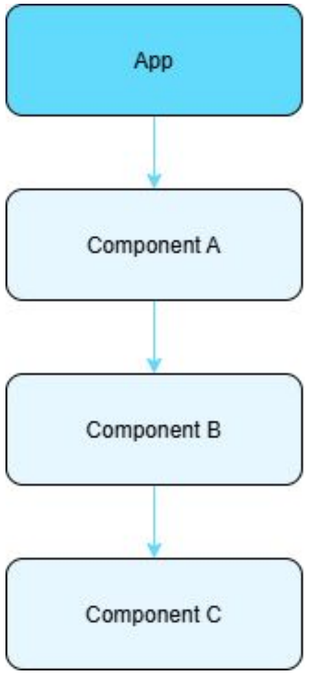


Data Flow in Our Project

State Flow in the Previous Projects

The Problem- Props Drilling

When State Goes Too Deep



Key Concept

- React uses one-way data flow
- State should live in the closest common parent
- This pattern is called *Lifting State Up*
- Children can update parent state via callback props

Form Handling & Validation in React

Forms are essential in web applications:

- Login
- Registration
- Checkout
- Profile update
- Contact forms

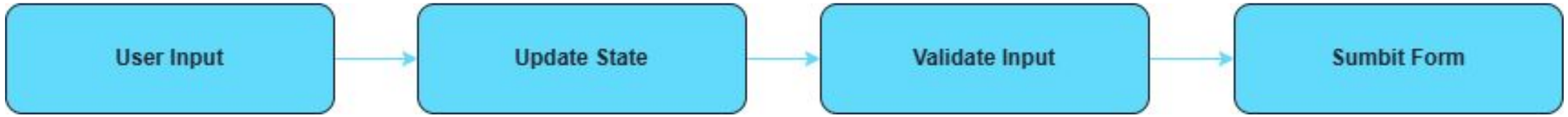
Challenges of Handling Forms

Challenges of Handling Forms

- managing multiple inputs
- storing form state
- writing validation logic
- displaying error messages
- handling submission

Basic Form Handling in React

- input state
- *onChange*
- *onSubmit*
- validation
- error state



Problems With Manual Forms

Manual form handling often leads to:

- lots of repetitive code
- difficult validation logic
- many state variables
- harder maintenance

Introducing React Hook Form

- minimal re-renders
- better performance
- simpler form management
- built-in validation support

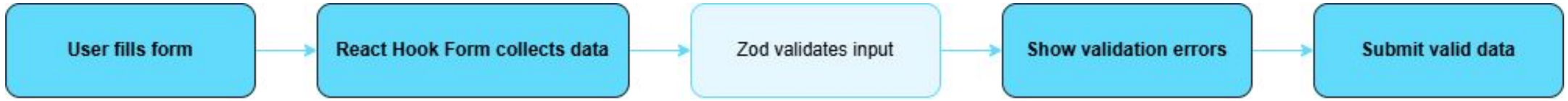
Schema Validation with Zod

A schema validation library used to define validation rules.

- structured validation
- reusable rules
- clear error messages

React Hook Form + Zod

npm install react-hook-form zod @hookform/resolvers



Best Practices for Form Validation

- validate required fields
- validate input format (email, password)
- show clear error messages
- prevent invalid submission
- keep validation rules consistent
- Prevent malicious scripts and potential injection attempts

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Thank You !